Cirata Floating Solar: modelled yield brief

192 MWp DC / 145 MWac, Cirata reservoir, West Java. Built 2026-10-20T13:51:06+00:00 with pvlib 0.16.1.

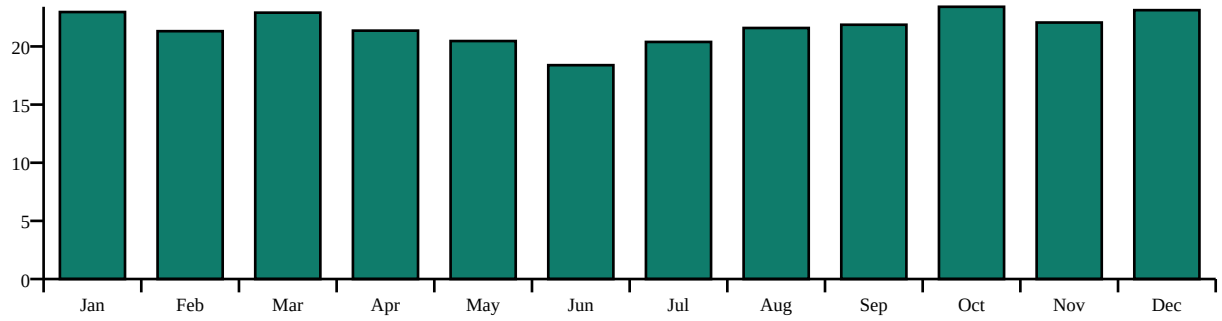
SYNTHETIC WEATHER. No real weather data was available for this build. Every number in this brief tests the pipeline only and is not an estimate of Cirata's output.

What this is. Cirata publishes only annual and cumulative output, so this is a forecast from a physics model checked against sparse reports. It is not a calibrated model and not live data.

Key figures (mid scenario unless a range is given)

Metric	Value	Range	Tag
Annual energy, P50	258 GWh	223-287 GWh (scenarios)	modelled
Weather spread (mid)	P90 256 / P10 267 GWh	12 weather years	modelled
Capacity factor (AC)	20.3%	17.6%-22.6%	modelled
Specific yield	1341 kWh/kWp	1163-1496	modelled
Performance ratio	84.4%		modelled
Revenue per year	US$15M	US$11-20M	assumed tariff
Build cost		US$100-130M	assumed
Homes powered	~172,000	93,000-239,000	assumed use
Avoided CO2 per year	219 kt	167-273 kt	assumed factor

Monthly energy, mid scenario, median across weather years (GWh)



Scoreboard: model vs reports (GWh)

Reported figures were not used to set any parameter. Each modelled value covers exactly the reporting period, repeated over every weather year; the envelope spans all scenarios. Grid curtailment is not modelled.

Reported figure	Period	Reported	Model mid P50	Envelope	Position	Reliability
Inauguration to June 2024	2023-11-09 to 2024-06-30	158	166.6	139-191	inside envelope	high
Cumulative to mid-Nov 2024	2023-11-09 to 2024-11-15	267	264.3	223-306	inside envelope	medium
Calendar 2024	2024-01-01 to 2024-12-31	279	257.5	220-299	inside envelope	medium
AGM 2025 figure (probably 2024)	2024-01-01 to 2024-12-31	273.68	257.5	220-299	inside envelope	medium
Claimed annual yield	annual	245-300	257.5	220-299	-	low

Assumptions (low = pessimistic for yield)

low/high set every assumed parameter to its pessimistic/optimistic end at once: a deliberately wide envelope, not a probability interval.

Assumption	Low	Mid	High	Unit	Tag
Module tilt	10	10	10	deg	assumed

Assumption	Low	Mid	High	Unit	Tag
Module azimuth	0	0	0	deg from N	assumed
Ground coverage ratio (row spacing)	0.75	0.65	0.55	-	assumed
Row width (module slant length)	2.3	2.3	2.3	m	assumed
Module height above water	0.5	0.5	0.5	m	assumed
Bifaciality factor	0.0	0.7	0.8	-	assumed
Share of surface under array that is float, not water	0.0	0.3	0.5	-	assumed
Float surface albedo	0.5	0.5	0.5	-	assumed
Module heat-loss coefficient (cooling)	29	35	50	W/m2K	assumed
Power temperature coefficient	-0.0037	-0.0034	-0.0029	1/degC	assumed
Soiling loss (dust, bird droppings)	0.04	0.02	0.01	fraction	assumed
Mismatch, DC wiring and wave motion	0.035	0.025	0.015	fraction	assumed
Light-induced and nameplate loss	0.02	0.01	0.003	fraction	assumed
Availability (uptime)	0.97	0.985	0.995	fraction	assumed
Inverter nominal efficiency	0.975	0.98	0.985	fraction	assumed
AC loss to export point (transformers, cables)	0.025	0.015	0.01	fraction	assumed
Module degradation	0.007	0.005	0.004	fraction/yr	assumed
Extra degradation from humid over-water environment	0.003	0.001	0.0	fraction/yr	assumed
Electricity sale price (PPA tariff)	0.05	0.058	0.07	US$/kWh	assumed
Capital cost	100	115	130	US$ million	assumed
Household electricity use	2400	1500	1200	kWh/yr	assumed
Java-Bali grid emission factor	0.75	0.85	0.95	tCO2/MWh	assumed
DC capacity		192000		kWp	reported
AC capacity (export limit)		145000		kW	reported
Location (dam point)		-6.70056, 107.36694		deg	reported
Altitude		220		m	assumed
Water surface type for albedo		green_water_ripples_up_to_2.5cm		-	assumed

Sources

PLN Nusantara Renewables statement: 158 GWh from inauguration to June 2024 (primary). URL not recorded in this beta; add before publishing.

Secondary press reports: 267 GWh cumulative to mid-Nov 2024; 279 GWh for 2024 (secondary). URLs not recorded in this beta; add before publishing.

2025 AGM report: 273.68 GWh (year not stated) (company report). URL not recorded in this beta; add before publishing.

Claimed annual yields of 245-300 GWh across press and company statements (claims). Expectations, not measurements.

192 MWp DC / 145 MWac; dam location; inauguration 9 Nov 2023 by PLN and Masdar; PowerChina EPC (verified by author). Supplied as verified by the project author; URLs to be added.

pvlib python (Holmgren, Hansen, Mikofski 2018, JOSS 3(29) 884) (software) - https://pvlib-python.readthedocs.io. Solar position (NREL SPA), transposition, infinite-sheds bifacial model, PVsyst cell temperature, PVWatts DC and inverter models.

Dvoracek & Hannabas 1990, inland water albedo vs solar elevation (as implemented in pvlib.albedo) (peer reviewed) - https://pvlib-python.readthedocs.io/en/stable/reference/generated/pvlib.albedo.inland_water_dvoracek.html. Water albedo model.

Dorenkamper et al. 2021, The cooling effect of floating PV in two different climate zones, Solar Energy 214 (peer reviewed). Basis for the heat-loss coefficient range; values recalled, not re-checked.

SYNTHETIC clear-sky fallback (not real weather) (synthetic placeholder). Files: none; generated in engine/weather.py